

11 Approximating Probabilities

- A probability is a theoretical long run relative frequency.
- The probability of event A can be approximated by simulating, according to the assumptions corresponding to the probability measure P , the random phenomenon a large number of times and computing the relative frequency of A .

$$P(A) \approx \frac{\text{number of repetitions on which } A \text{ occurs}}{\text{number of repetitions}}, \quad \text{for a large number of } P\text{-repetitions}$$

- In practice, many repetitions of a simulation are performed on a computer to approximate what happens in the “long run”. However, we often start by carrying out a few repetitions, (1) by hand to help make the process more concrete, or (2) by computer to check that our code is working correctly
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency*.
- Use a margin of error of $2/\sqrt{N}$ when approximating many probabilities based on a single simulation.
- Be careful when conditioning, especially when conditioning on values of continuous random variables.

Example 11.1. Roll a fair four-sided die twice and let X be the sum and Y the larger of the two rolls.

1. Explain how you would use simulation to approximate
 - a. $P(X = 6)$

- b. $P(X = 6, Y = 4)$
2. In 10000 repetitions of the simulation, $X = 6$ on 1915 repetitions. Approximate $P(X = 6)$, including a margin of error.
 3. Explain how you would use simulation to approximate the distribution of X . What would the margin of error be?

Example 11.2. Suppose that Regina and Cady's arrival times (minutes after noon) are independent, with Regina's arrival time R uniformly distributed between 0 and 60, and Cady's Y arrival time following a Normal(30, 10) distribution. Let $W = |R - Y|$.

1. Explain in full detail how you could conduct an appropriate simulation using spinners and use the results to approximate the probability $P(W < 15)$.
2. In 10000 simulated repetitions, $W < 15$ in 4975. Approximate and interpret $P(W < 15)$; include a margin of error.

Example 11.3. Consider simulating a randomly selected U.S. adult and determining whether or not the person is age 18-29 and whether or not they use Snapchat. Let A be the event that

the selected adult is age 18-29, C be the event that the selected adult uses Snapchat, and P correspond to randomly selecting an American adult. Suppose that $P(A) = 0.20$, $P(C) = 0.24$, and $P(A \cap C) = 0.13$.

1. In 10000 repetitions of an appropriate simulation

- A occurs on 1989 repetitions
- C occurs on 2432 repetitions
- Both A and C occur on 1309 repetitions.

Use the simulation results to approximate $P(C|A)$.

2. What is the margin of error for your estimate in the previous part?

3. What is another method for performing the simulation and estimating $P(C|A)$ that has a margin of error of 0.01? What is the disadvantage of this method?

Example 11.4. We'll continue with the dice rolling example: roll two fair four-sided dice and let X be the sum and Y the larger of the rolls. For each of the following, describe how to approximate the probability with a margin of error of 0.01.

1. $P(X = 6|Y = 4)$.

2. $P(Y = 4|X = 6)$.

Example 11.5. Suppose that Regina and Cady's arrival times (minutes after noon) are independent, with Regina's arrival time R uniformly distributed between 0 and 60, and Cady's Y arrival time following a Normal(30, 10) distribution. We want to approximate the probability that they arrive within 15 minutes of each given that Regina arrives at 12:40. Explain in detail how you would conduct a simulation to approximate this probability. (Be careful!)